PRODUCT PROPOSAL

Semester 5 Integrated PBL — Information Technology

THRIFTY

Location-Based Secondhand Fashion Marketplace with AI-Powered Clothing Condition Grading

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1. Cover and Team Identity

1.1 Product Title

THRIFTY: Location-Based Secondhand Fashion Marketplace with AI-Powered Clothing Condition Grading

1.2 Team Identity

Name	Role
Tiara Febrianie	Team Lead & Lead Developer
Joseph Atem Deng Aruei	ML Engineer / System Architect
Andhika Daffa	Mobile Developer
Garino Wijaya	UI/UX Developer
Raihan Zaky	QA / Tester

1.3 Integrated Courses and Lecturers

Course	Instructor	Class
Mobile Programming	Habibie Ed Dien, S.Kom., M.T.	TI - 3I
Machine Learning	Muhammad Afif Hendrawan, S.Kom., M.T.	TI - 3I
Software Quality Assurance	M. Hasyim Ratsanjani, S.Kom., M.Kom.	TI - 3I

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2. Executive Summary

Secondhand fashion provides an opportunity to extend the useful life of clothing while offering consumers more accessible purchasing options. However, finding secondhand products near a user's location can be difficult, since product listings are often scattered across different platforms and clothing condition descriptions vary between sellers. Thrifty is a mobile-based secondhand fashion marketplace that helps users discover nearby clothing products through location-based search and a customizable search radius. The application also integrates a machine learning feature that analyzes product images and classifies the visible condition of clothing into four grades: Grade A, Grade B, Grade C, and Grade D. The initial model focuses on four clothing categories: shirts, T-shirts, pants, and jackets. Thrifty presents product images, prices, seller information, location, and AI-generated condition grades within a single application. The project will be developed using Flutter and evaluated through functional, usability, compatibility, performance, and machine learning testing. The final target is a functional mobile prototype in which the AI condition-classification model achieves a minimum test accuracy of 80%.

3. Background and Problem Formulation

3.1 Background

Secondhand fashion has become an increasingly relevant alternative for consumers who want to purchase clothing at more accessible prices while giving previously owned items an opportunity to be reused. The growing number of thrift stores, individual sellers, and online secondhand marketplaces has made it easier for consumers to access used clothing. However, the process of finding suitable secondhand products can still be inefficient, particularly for users who prefer products that are available close to their own location.

Secondhand product listings are commonly shared through social media, messaging platforms, and general marketplace applications that are not specifically designed for this market. These platforms allow sellers to display their products, but buyers often need to browse through many unrelated listings before finding items that are actually located nearby. A dedicated location-based marketplace can simplify this process by displaying products according to proximity and allowing users to define the maximum distance they are willing to search within.

Another recurring challenge is determining the condition of secondhand clothing before purchase. Product quality is usually communicated only through seller-written descriptions and photographs, and because different sellers tend to describe similar conditions in different ways, buyers often find it difficult to compare products consistently. A standardized, image-based grading system can provide buyers with an additional, more objective reference when evaluating secondhand clothing.

Thrifty addresses these two problems together by combining location-based product discovery with an AI-powered clothing condition grading feature. Users can search for secondhand products near their location using an adjustable search radius, while sellers receive an AI-generated condition grade automatically whenever they upload product images. At this stage, the system focuses on four initial clothing categories, namely shirts, T-shirts, pants, and jackets, which were selected because they represent common and frequently resold secondhand clothing items.

A mobile application is a practical form factor for this solution because most buyers and sellers browse and manage marketplace listings from their phones, often while on the move, and location-based features depend on the device's built-in positioning capability. Machine learning is relevant here because manually grading every uploaded image at scale is impractical for sellers, while a trained image-classification model can provide a fast and consistent first assessment of visible clothing condition as soon as a photo is submitted.

3.2 Problem Formulation

1. How can users discover secondhand fashion products that are available near their location more efficiently?
2. How can users search for secondhand products according to a preferred distance range from their current location?
3. How can the condition of secondhand clothing be presented to buyers using a standardized classification system?
4. How can machine learning be used to classify the visible condition of secondhand clothing directly from product images?
5. How can location-based product discovery and AI-powered condition grading be integrated into a single, practical mobile application?

3.3 Technical and Data Feasibility Study

The main machine learning problem addressed by Thrifty is image classification for secondhand clothing condition. The initial model is scoped to four clothing categories and four condition grades, as summarized below.

Category	Description
Shirt	Formal and casual shirts
T-Shirt	Casual and everyday T-shirts
Pants	Jeans, trousers, and similar bottoms
Jacket	Casual jackets and outerwear

Grade	Description
A — Like New	Clothing appears almost new, with little to no visible sign of use
B — Very Good	Clothing shows minor visible signs of use but remains in very good condition
C — Good / Usable	Clothing shows visible signs of use or minor defects but remains fully usable
D — Significant Wear	Clothing shows noticeable visible defects or substantial signs of wear

The dataset for the initial model will be assembled from suitable, publicly available clothing image sources, supplemented where necessary with additional images collected by the team during development. Dataset suitability will be assessed based on image quality, coverage of the four target categories, availability of condition labels, class balance, and similarity to the kind of images sellers are expected to upload in practice. Because an exact dataset size has not yet been finalized, the team will treat Weeks 1–3 as a dedicated dataset exploration and feasibility phase, during which the final dataset source, approximate size, and class distribution will be determined.

The initial modeling approach is transfer learning using MobileNetV2, chosen for its lightweight architecture and suitability for mobile-oriented deployment. The planned data split is approximately 80% training, 10% validation, and 10% testing, to be adjusted if the finalized dataset size or class balance requires it. The main evaluation target is a minimum of 80% test accuracy, supported by precision, recall, F1-score, and a confusion matrix.

Given a 16-week project timeline, this scope is considered realistic: the first three weeks are allocated to dataset feasibility, model development and baseline training take place from Week 5 through Week 12, and the remaining weeks are reserved for optimization, mobile integration, and testing. The single, well-defined ML task — four categories, four grades, one classification model — keeps the machine learning workload proportionate to the time and computational resources available to a student team.

4. Project Objectives and Scope

4.1 Project Objectives

1. Develop a mobile-based secondhand fashion marketplace application using Flutter.
2. Implement location-based product discovery that allows users to find nearby secondhand clothing.
3. Implement a customizable search radius for nearby product discovery.
4. Develop an image-classification model that classifies the visible condition of clothing into Grade A–D.
5. Integrate the machine learning model into the mobile application so that predictions are generated in real time.
6. Provide structured product information, including images, prices, categories, seller information, condition grades, and location.

7. Provide sellers with a practical listing workflow supported by AI-based condition grading.
8. Evaluate the application's functionality and usability through documented software testing.
9. Evaluate the machine learning model using measurable performance metrics.
10. Achieve a minimum test accuracy of 80% for the clothing condition classification model.

4.2 Scope

Included / Developed

- User registration and login
- User profile
- Buyer and seller functionality
- Product listing and product image upload
- Clothing category selection (Shirt, T-Shirt, Pants, Jacket)
- AI-powered clothing condition grading (Grade A–D)
- Location-based product discovery
- Search radius selection and nearby product browsing
- Product search and filtering
- Product detail page (image, price, category, condition, seller, location)
- Listing management for sellers
- Machine learning model evaluation
- Functional and non-functional software testing

Not Included

- Automated payment gateway
- Integrated delivery management
- Brand authenticity verification
- AI-based fashion or outfit recommendations
- Price prediction
- Clothing categories outside the four initial categories
- Advanced marketplace management features beyond the core prototype
- Large-scale commercial deployment

4.3 Stakeholders

Stakeholder	Interest
Buyers	Find nearby secondhand clothing and understand the visible condition of products before purchase
Individual Sellers	List secondhand clothing and reach potential buyers near their location
Thrift Businesses / UMKM	Promote products locally and reach more nearby customers
Project Team	Design, develop, integrate, test, and evaluate the Thrifty application
Course Lecturers	Assess the integration of Mobile Programming, Machine Learning, and Software Quality Assurance
Dataset Providers	Provide clothing image data that supports machine learning development

4.4 Assumptions and Limitations

Assumptions

- Users are able to provide reasonably clear clothing images.
- A suitable image dataset is available for the four selected categories.
- Users grant the required location permission when using nearby-product discovery.
- Sellers provide accurate product descriptions and prices.
- Android devices are available to the team for development and testing.

Limitations

- The initial model supports only four clothing categories: shirts, T-shirts, pants, and jackets.
- The model evaluates the visible condition of clothing from images only.
- Hidden defects, hygiene, material quality, and authenticity cannot be assessed by the model.
- Poor image quality, blur, or insufficient lighting may reduce classification accuracy.
- Location functionality depends on the device's location service being available and enabled.
- The project is limited to the 16-week PBL development period.
- The final product is a functional academic prototype rather than a full commercial marketplace.

4.5 Success Criteria

1. The Thrifty mobile application installs and runs successfully on an Android device.
2. Users can register, log in, and manage a basic profile.
3. Sellers can create, edit, and manage secondhand product listings.
4. Buyers can discover and browse nearby secondhand products within a selected search radius.
5. Product detail pages display image, price, category, seller information, location, and condition grade.
6. The machine learning model produces a real Grade A–D prediction inside the application.
7. The model achieves a minimum test accuracy of 80%, supported by a documented confusion matrix.
8. Precision, recall, and F1-score are reported alongside accuracy.
9. Functional testing is executed and documented across the core features.
10. At least one non-functional testing type (usability, compatibility, or performance) is executed and documented.
11. The primary buyer and seller flows operate without critical errors.

5. Description of the Proposed Product

5.1 Overview of the Solution

Thrifty is a mobile-based secondhand fashion marketplace that connects buyers and sellers through location-based product discovery. When a user opens the application and grants location access, Thrifty displays secondhand clothing available nearby and allows the user to narrow or widen the results using an adjustable search radius. From this list, the user can open any product to view its image, price, category, seller information, location, and AI-generated condition grade.

On the seller side, creating a listing begins with uploading one or more images of the clothing item. The uploaded image is automatically processed by the machine learning model, which classifies its visible condition into Grade A–D. The seller reviews this grade, adds the remaining product details such as price, description, category, and location, and then publishes the listing. The predicted grade becomes part of the product information that buyers see when browsing.

In this way, the AI clothing-condition grading feature is not a separate or optional add-on; it is embedded directly in the core listing flow, so that every published product carries a standardized, image-based condition indicator alongside the location information that helps buyers find it nearby.

5.2 Users and Usage Scenarios

Buyer

The buyer opens Thrifty and grants location access, after which the application displays secondhand products available nearby. The buyer can select a clothing category and set a preferred search radius to narrow the results, then open a specific product to review its image, price, category, seller information, location, and condition grade before deciding whether to proceed.

Buyer Flow: Open Thrifty → Allow Location Access → Select Category → Set Search Radius → Browse Nearby Products → Select Product → View Product Details → Review Condition Grade

Seller

The seller logs into Thrifty and starts a new product listing by uploading an image of the clothing item and selecting its category. The image is submitted for condition analysis, and the machine learning model returns a predicted Grade A–D result. The seller reviews the grade, completes the remaining product information such as price, description, and location, and publishes the listing.

Seller Flow: Login → Add Product → Upload Image → Select Category → AI Condition Grading → Review Grade → Enter Product Information → Add Location → Publish Listing

5.3 Main Features

Feature	Description	ML
AI Clothing Condition Grading	Classifies the visible condition of clothing into Grade A–D from an uploaded product image	[ML]
Location-Based Product Discovery	Helps users find secondhand products available near their location	
Search Radius	Allows users to define the distance range used for nearby product discovery	
Nearby Product Browsing	Displays products ordered or filtered by their distance from the user	
Product Listing	Allows sellers to upload clothing images and enter product information	
Product Detail	Displays product image, price, category, condition grade, seller, and location	
Search and Filter	Allows users to filter products by category, price, condition, and distance	
Authentication	Provides user registration and login functionality	
User Profile	Displays and manages basic user account information	
Listing Management	Allows sellers to add, edit, and manage their own product listings	

The clothing-condition classifier is the single main ML feature of the application; location-based discovery, listing, search, filtering, and account management serve as the supporting application features that surround it.

6. Initial Technical Design

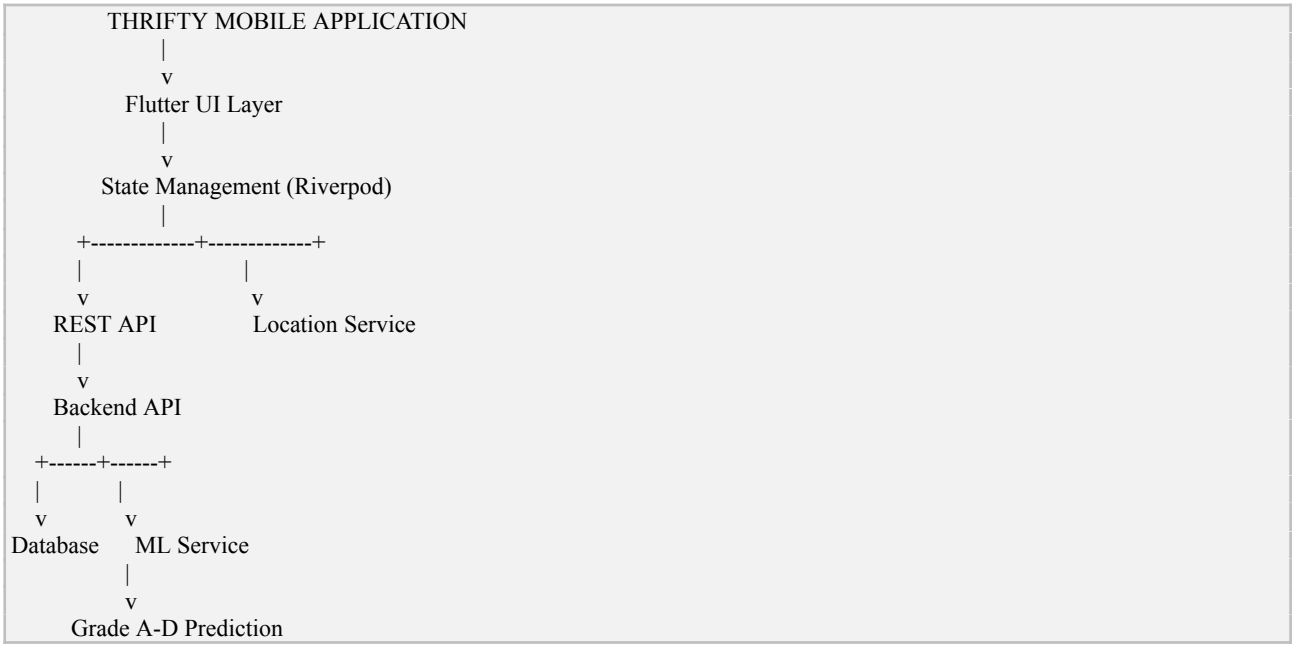
6.1 Mobile Application Architecture (Flutter)

Thrifty will be developed using Flutter. The initial application is planned to include the following screens:

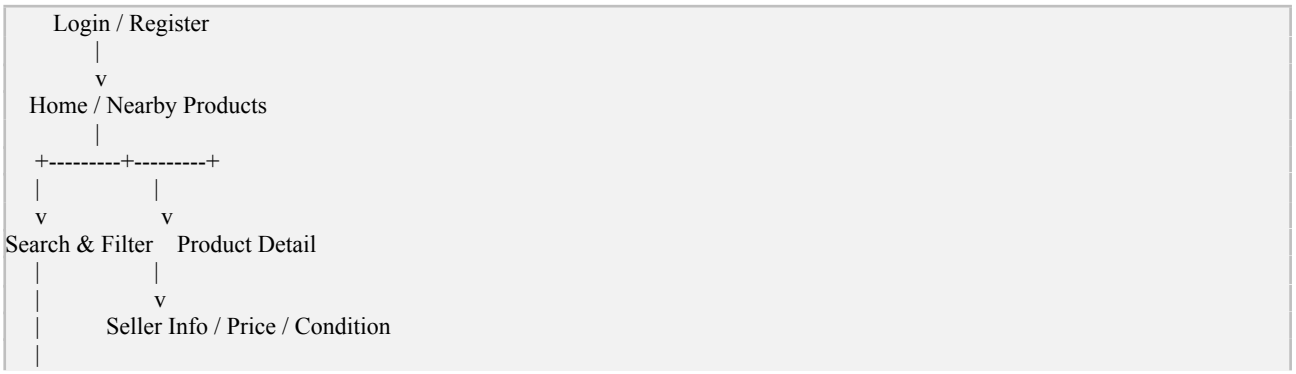
- Login
- Register
- Home / Nearby Products
- Search and Filter
- Product Detail
- Add Product
- AI Grading Result
- My Listings
- Profile

State management will be handled using Riverpod, covering authentication state, product data, user location, active filters, and machine learning prediction results. Application data such as product listings and user accounts will be managed through a backend REST API and database, while the on-device location service supplies the user's coordinates for nearby product discovery.

Proposed Architecture



Application Navigation



```
  v
My Listings ----> Add Product ----> Upload Image
                                       |
                                       v
                                AI Condition Grading
                                       |
                                       v
                                  Grade A - D
                                       |
                                       v
                                 Publish Product
```

6.2 Dataset and Machine Learning Methodology

6.2.1 ML Task

The ML feature is formulated as an image classification task. Input: a photograph of a secondhand clothing item. Output: one condition grade among Grade A, Grade B, Grade C, or Grade D.

6.2.2 Categories and Condition Classes

The initial model covers four clothing categories (Shirt, T-Shirt, Pants, Jacket) and the four condition grades defined in Section 3.3.

6.2.3 Dataset Plan

The dataset will consist of clothing images drawn from suitable, publicly available sources, supplemented by additional images collected by the team if needed. The dataset will be assessed based on the number of images, category distribution, condition-label distribution, image quality, label consistency, and similarity to expected real-world seller uploads. The exact dataset source and final image count will be confirmed during the Week 1–3 feasibility study rather than fixed in advance.

6.2.4 Data Preprocessing

- Image collection
- Data cleaning and duplicate removal
- Label verification
- Image resizing and normalization
- Data augmentation
- Dataset splitting

6.2.5 Training Methodology

The proposed approach is transfer learning using MobileNetV2, selected for its balance between accuracy and suitability for mobile deployment.

```
    Clothing Image Dataset
            |
       Data Cleaning
            |
     Label Verification
            |
       Preprocessing
            |
Train / Validation / Test Split
            |
      Data Augmentation
            |
     Transfer Learning
            |
        MobileNetV2
            |
         Training
```

```
        |
   Evaluation
        |
   Optimization
        |
Mobile Integration
```

6.2.6 Data Split

The initial dataset split will be approximately 80% training, 10% validation, and 10% testing, to be adjusted if necessary once the final dataset size and class distribution are known.

6.2.7 Evaluation Metrics

The model will be evaluated using accuracy, precision, recall, F1-score, and a confusion matrix, with class-level results reviewed to identify whether any single condition grade is more difficult to classify than the others. The initial target is a minimum test accuracy of 80%.

6.2.8 Application Integration

After evaluation and optimization, the selected model will be integrated into the Thrifty application so that a seller's uploaded image produces a real, on-screen Grade A–D classification result.

7. Initial QA Plan

Testing Scope

The initial QA process covers registration and login, product listing, image upload, AI clothing condition grading, location-based product discovery, search radius, product search and filtering, product detail, and listing management.

Testing Types

Functional Testing

Functional testing verifies that each core feature behaves according to its requirements, including registration, login, listing creation, image upload, AI grading output, location access, nearby-product results, search, filtering, product detail display, and listing management.

Non-Functional Testing

Non-functional testing will cover usability (navigation clarity, ease of adjusting the search radius, and ease of understanding the Grade A–D result), compatibility (behavior across different Android devices and screen sizes), and performance (product loading time, location retrieval time, and ML prediction response time).

ML Testing

ML testing evaluates accuracy, precision, recall, F1-score, and the confusion matrix on unseen test images, including images captured under varying quality conditions such as different lighting and backgrounds.

Testing Schedule

Week	Activity
9–10	Test case preparation
12	Interim testing
15	Final testing

8. Work Plan and Team Organization

8.1 Schedule and Milestones

Week	Main Activities	Milestone
1–3	Problem analysis, requirements gathering, dataset exploration, technical feasibility study	—
4	Proposal finalization and initial QA plan	CP-1 Proposal
5–6	UI/UX design, database design, Flutter project setup, dataset preparation	—
7–8	Baseline ML model training, backend development, core mobile features	CP-2 Milestone 1
9–10	Model improvement, API integration, location-based discovery, product features	—
11–12	ML integration, search/filtering, product details, interim testing	CP-3 Milestone 2
13–14	Model optimization, UI refinement, compatibility and usability testing	—
15	Final testing, bug fixing, model evaluation, QA documentation	—
16	Final evaluation, documentation, and product demonstration	CP-4 Final

8.2 Division of Roles and Responsibilities

Tiara Febrianie — Team Lead & Lead Developer

- Coordinate the overall project and manage the development schedule
- Supervise implementation and lead application development
- Coordinate integration between the mobile application, backend, and ML components
- Ensure the project remains aligned with the agreed scope

Joseph Atem Deng Aruei — ML Engineer / System Architect

- Conduct dataset exploration and preprocessing
- Train, evaluate, and optimize the clothing-condition classification model
- Design the initial system architecture
- Support ML integration into the mobile application

Andhika Daffa — Mobile Developer

- Develop the Flutter application, screens, and navigation
- Implement product browsing and product listing functionality
- Integrate application features with the backend API
- Support integration of ML results into the app UI

Garino Wijaya — UI/UX Developer

- Design the user interface and user flow
- Develop and maintain interface components and visual consistency
- Improve usability and support usability evaluation

Raihan Zaky — QA / Tester

- Prepare the QA plan and functional test cases

- Execute functional, usability, compatibility, and performance testing
- Support ML testing and record defects
- Prepare QA documentation and test reports

9. Industrial Partner Plan

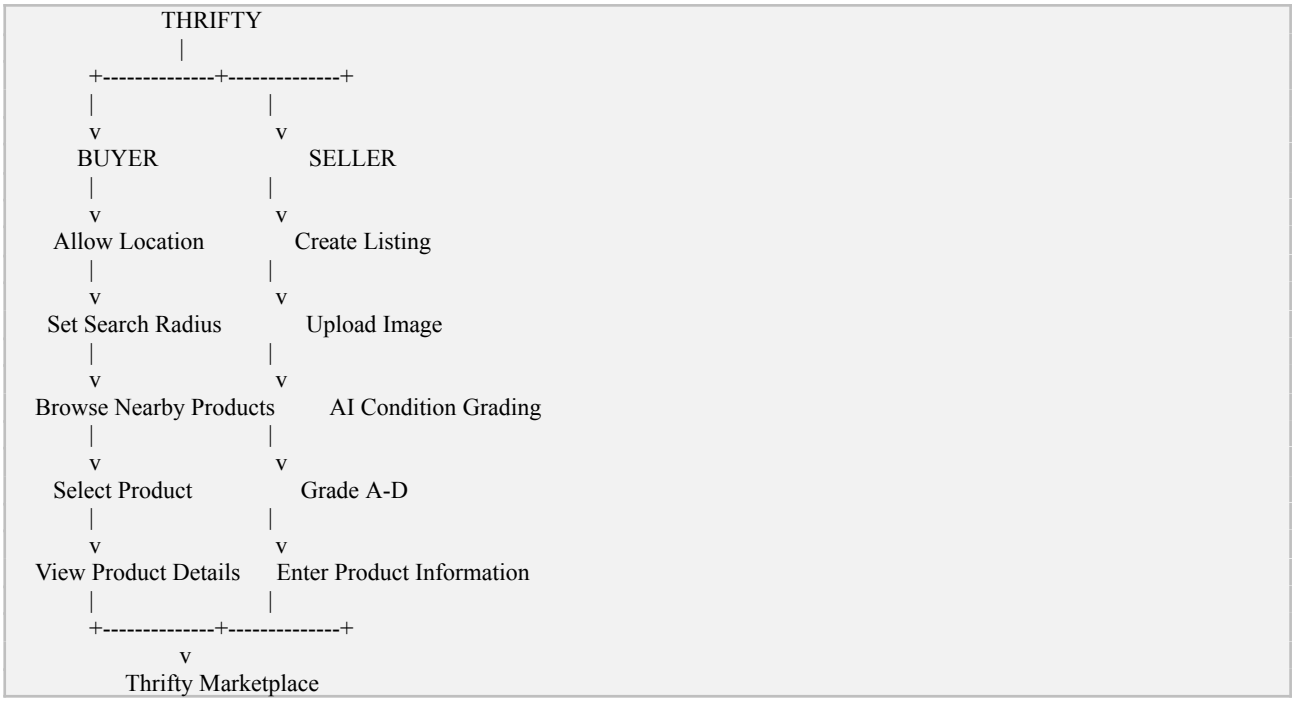
No industry partner at the proposal stage. The project will initially rely on publicly available clothing image datasets and additional images collected by the development team when necessary. Potential local secondhand sellers may be considered later for usability feedback or informal validation during development, but no formal industry partner has been established at this stage.

10. Appendices

Appendix A. Initial Application Screens

- Login
- Register
- Home / Nearby Products
- Search and Filter
- Product Detail
- Add Product
- AI Grading Result
- My Listings
- Profile

Appendix B. Main User Flow



Appendix C. Machine Learning Workflow

```
Dataset
  |
Data Cleaning
  |
Label Verification
  |
Preprocessing
  |
Dataset Splitting
  |
Data Augmentation
  |
Transfer Learning
  |
MobileNetV2
  |
Training
  |
Model Evaluation
  |
Model Optimization
  |
Mobile Application Integration
```

Appendix D. Clothing Condition Grading Reference

Grade	Condition
Grade A	Like New
Grade B	Very Good
Grade C	Good / Usable with Visible Wear
Grade D	Significant Wear / Visible Defects

Appendix E. Initial QA Test Categories

Functional Testing

- Authentication
- Product listing
- Image upload
- AI grading
- Nearby product discovery
- Search radius
- Product search and filtering
- Product detail
- Listing management

Non-Functional Testing

- Usability
- Compatibility
- Performance

ML Evaluation

- Accuracy
- Precision

- Recall
- F1-score
- Confusion matrix
- Unseen-image testing

Appendix F. Initial Technical Architecture

